# Supplementary File for Paper: IStacking: An Iterative Stacking Model to Solve Imbalanced Data Classification Problems

This supplementary provides detailed information about the 65 datasets we used, as well as all the symbols and acronyms employed in the text.

Table A gives the description of the 65 datasets. Imbalanced datasets feature with imbalance rate (IR), number of samples, and number of features. IR is the ratio of majority samples to minority ones. The collected datasets are sorted by IR from small to large.

TABLE A
DATASET DESCRIPTION.

| Name | IR | Number of samples | Number of features |
|---|---|---|---|
| glass1 | 1.82 | 214 | 7 |
| ecoli-0_vs_1 | 1.86 | 220 | 7 |
| wisconsin | 1.86 | 683 | 9 |
| pima | 1.87 | 768 | 8 |
| iris0 | 2.00 | 150 | 4 |
| glass0 | 2.06 | 214 | 9 |
| yeast1 | 2.46 | 1484 | 8 |
| haberman | 2.78 | 306 | 3 |
| vehicle2 | 2.88 | 846 | 18 |
| vehicle1 | 2.90 | 846 | 18 |
| vehicle3 | 2.99 | 846 | 18 |
| glass-0-1-2-3_vs_4-5-6 | 3.20 | 214 | 9 |
| vehicle0 | 3.25 | 846 | 18 |
| ecoli1 | 3.36 | 336 | 7 |
| new-thyroid1 | 5.14 | 215 | 5 |
| new-thyroid2 | 5.14 | 215 | 5 |
| ecoli2 | 5.46 | 336 | 7 |
| segment0 | 6.02 | 2308 | 19 |
| glass6 | 6.38 | 214 | 9 |
| yeast3 | 8.10 | 1484 | 8 |
| ecoli3 | 8.60 | 336 | 7 |
| ecoli-0-3-4_vs_5 | 9.00 | 200 | 7 |
| yeast-2_vs_4 | 9.08 | 514 | 8 |
| ecoli-0-6-7_vs_3-5 | 9.09 | 222 | 7 |
| ecoli-0-2-3-4_vs_5 | 9.10 | 202 | 7 |
| yeast-0-3-5-9_vs_7-8 | 9.12 | 506 | 8 |
| yeast-0-2-5-7-9_vs_3-6-8 | 9.14 | 1004 | 8 |
| yeast-0-2-5-6_vs_3-7-8-9 | 9.14 | 1004 | 8 |
| ecoli-0-4-6_vs_5 | 9.15 | 203 | 6 |
| ecoli-0-1_vs_2-3-5 | 9.17 | 244 | 7 |
| ecoli-0-2-6-7_vs_3-5 | 9.18 | 224 | 7 |
| glass-0-4_vs_5 | 9.22 | 92 | 9 |
| ecoli-0-3-4-6_vs_5 | 9.25 | 205 | 7 |
| ecoli-0-3-4-7_vs_5-6 | 9.28 | 257 | 7 |
| vowel0 | 9.98 | 988 | 13 |
| ecoli-0-6-7_vs_5 | 10.00 | 220 | 6 |
| glass-0-1-6_vs_2 | 10.29 | 192 | 9 |
| ecoli-0-1-4-7_vs_2-3-5-6 | 10.59 | 336 | 7 |
| led7digit-0-2-4-5-6-7-8-9_vs_1 | 10.97 | 443 | 7 |
| ecoli-0-1_vs_5 | 11.00 | 240 | 6 |
| glass-0-1-4-6_vs_2 | 11.06 | 205 | 9 |
| ecoli-0-1-4-7_vs_5-6 | 12.28 | 332 | 6 |
| cleveland-0_vs_4 | 12.62 | 177 | 13 |
| ecoli-0-1-4-6_vs_5 | 13.00 | 280 | 6 |
| shuttle-c0-vs-c4 | 13.87 | 1829 | 9 |
| yeast-1_vs_7 | 14.30 | 459 | 7 |
| glass4 | 15.47 | 214 | 9 |

| | | | |
|---|---|---|---|
| ecoli4 | 15.80 | 336 | 7 |
| page-blocks-1-3_vs_4 | 15.86 | 472 | 10 |
| dermatology-6 | 16.90 | 358 | 34 |
| glass-0-1-6_vs_5 | 19.44 | 184 | 9 |
| shuttle-6_vs_2-3 | 22.00 | 230 | 9 |
| yeast-1-4-5-8_vs_7 | 22.10 | 693 | 8 |
| glass5 | 22.78 | 214 | 9 |
| yeast-2_vs_8 | 23.10 | 482 | 8 |
| winequality-red-4 | 29.17 | 1599 | 11 |
| poker-9_vs_7 | 29.50 | 244 | 10 |
| yeast-1-2-8-9_vs_7 | 30.57 | 947 | 8 |
| winequality-white-9_vs_4 | 32.60 | 168 | 11 |
| yeast5 | 32.73 | 1484 | 8 |
| winequality-red-8_vs_6 | 35.44 | 656 | 11 |
| ecoli-0-1-3-7_vs_2-6 | 39.14 | 281 | 7 |
| yeast6 | 41.40 | 1484 | 8 |
| winequality-white-3-9_vs_5 | 58.28 | 1482 | 11 |
| shuttle-2_vs_5 | 66.67 | 3316 | 9 |

Table B lists all symbols that appear in the text.

TABLE B
ALL SYMBOLS.

| Name | Symbols |
|---|---|
| Training set | $\Omega$ |
| Importance of each partition | $I = \{I_1, I_2, \dots, I_n\}$ |
| New imbalanced dataset | $\widehat{\Omega}$ |
| Denoised dataset | $\breve{\Omega}$ |
| The number of subsets into which $\Omega$ is divided | $k_i$ |
| The subsets formed after $\Omega$ is split | $\{\Omega^1, \Omega^2, \dots, \Omega^{k_i}\}$ |
| The number of base classifiers | $n$ |
| Base classifiers | $C = \{c_i\}_{i=1}^n$ |
| The number of classifiers that correctly classify | $\alpha$ |
| The number of samples belonging to the same partition | $\beta$ |
| Average distance among samples belonging to the same partition | $d$ |
| Floor function | $\lfloor\ \rfloor$ |
| $\beta$ arctangent normalization | $\tilde{\beta}$ |
| $d$ arctangent normalization | $\tilde{d}$ |
| Change in $f(x)$ | $\Delta$ |
| Weight of each partition | $w = \{w_1, w_2, \dots, w_n\}$ |
| The number of majority samples in $\Omega$ | $M$ |
| The number of minority samples in $\Omega$ | $N$ |
| A synthetic minority sample | $\hat{x}$ |
| A minority point under consideration | $x_m$ |
| A randomly selected nearest neighbor | $x_n$ |
| A random number | $\delta$ |
| The number of subsets into which $\widehat{\Omega}$ is divided | $k_d$ |
| The subsets formed after $\widehat{\Omega}$ is split | $\{\widehat{\Omega}^1, \widehat{\Omega}^2, \dots, \widehat{\Omega}^{k_d}\}$ |
| The number of subsets into which $\breve{\Omega}$ is divided | $k$ |
| Test set | $\Phi$ |
| The training set formed by the prediction results of the first layer in Stacking | $P_\Omega = \{P_\Omega^1, P_\Omega^2, \dots, P_\Omega^n\}$ |
| The test set formed by the prediction results of the first layer | $P_\Phi = \{P_\Phi^1, P_\Phi^2, \dots, P_\Phi^n\}$ |
| A meta-classifier of the second layer in Stacking | $MC$ |
| Final results | $P_f$ |
| Data size input to base classifiers | $S$ |
| The number of features input to base classifiers | $F$ |
| Data size input to meta-classifier | $S_m$ |
| The number of features input to meta-classifier | $F_m$ |
| Computational complexity of the base classifiers | $O = \{O_1, O_2, \dots, O_n\}$ |
| Computational complexity of the meta-classifier | $O_m$ |
| The number of true positive samples | $T_P$ |

| | |
|---|---|
| The number of true negative samples | $T_N$ |
| The number of false positive samples | $F_P$ |
| The number of false negative samples | $F_N$ |
| The area under a Receiver Operating Characteristic curve | $AUC$ |
| F1 score | $F1$ |
| Precision value | $P_r$ |
| Recall value | $R_e$ |
| Gmean score | $\bar{G}$ |
| Specificity value | $S_P$ |

Table C lists all acronyms that appear in the text.

TABLE C
ALL ACRONYMS.

| Name | Acronyms |
|---|---|
| Synthetic Minority Oversampling TEchnique | SMOTE |
| Iterative Stacking | IStacking |
| Knowledge Extraction based on Evolutionary Learnin | KEEL |
| Importance Quantification Algorithm | IQA |
| Imbalance Rate | IR |
| Decision Tree | DT |
| MultiLayer Perceptron | MLP |
| k Nearest Neighbor | kNN |
| Support Vector Machine | SVM |
| Naive Bayes | NB |
| Self-Paced Ensemble Classifier | SPEC |
| Balance-Cascade Classifier | BCC |
| Balanced Random Forest Classifier | BRFC |
| EasyEnsemble Classifie | EEC |
| RUSBoost Classifier | RBC |
| UnderBagging Classifier | UBC |
| OverBoost Classifier | OBC |
| SMOTEBoost Classifier | SBC |
| OverBagging Classifier | OBaC |
| AdaCost Classifier | ACC |
| AdaUBoost Classifier | AUBC |
| Asymmetric Boosting Classifier | ABC |
| Compatible AdaBoost Classifier | CABC |
| Compatible Bagging Classifier | CBaC |